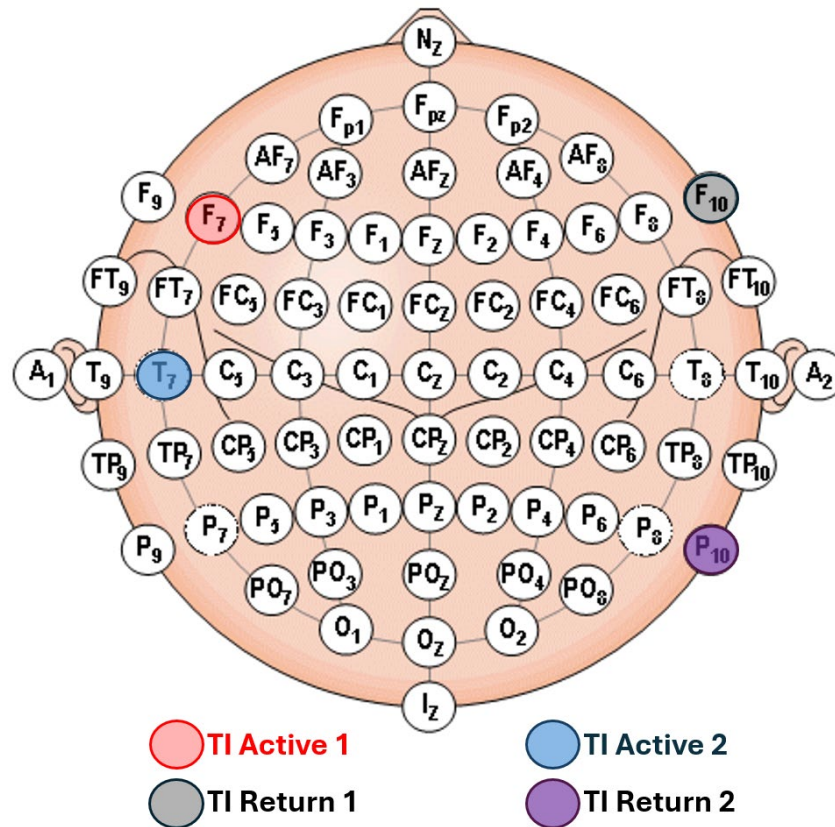


TI Stimulation Human Deployment

Setup: Electrode Placement



Find and mark halfway along the nasion-inion axis.

Find and mark halfway along the pre-auricular to opposite pre-auricular axis. The intersection between this and the previous mark is the vertex, Cz.

Find and mark the TI stimulation electrodes using the 10-10 positioning system.

Electrode Contact Test

Stimulation Parameters:

I1	
Frequency (Hz)	2000
Amplitude (mA)	0.5
I2 OFF	
Frequency (Hz)	N/A
Amplitude (mA)	N/A

Make ramp-time sufficient to minimise sensation (~5s) and stimulation time long enough to record voltage/impedance (~5-10s).

Record the measurement voltage (target: $\ll$ Compliance Voltage)

Calculate the load Resistance from recorded voltage and set current amplitude (target: $< 5k\Omega$)

Stimulation Parameters:

I1 OFF	
Frequency (Hz)	N/A
Amplitude (mA)	N/A
I2	
Frequency (Hz)	2000
Amplitude (mA)	0.5

Make ramp-time sufficient to minimise sensation (~5s) and stimulation time long enough to record voltage/impedance (~5-10s).

Record the measurement voltage (target: $\ll$ Compliance Voltage)

Calculate the load Resistance from recorded voltage and set current amplitude (target: $< 5k\Omega$)

Participant Sensation Test

Stimulation Parameters:

I1	
Frequency (Hz)	2000
Amplitude (mA)	0.5
I2 OFF	
Frequency (Hz)	2005
Amplitude (mA)	0.5

Make ramp-time sufficient to minimise sensation (~5s) and stimulation time long enough to record voltage/impedance (~5-10s).

Ask Participant which applies (record the selection):

No Sensation
Mild sensation (tolerable)
Painful or Significant discomfort

Mitigate if painful or significant discomfort

- Press problematic electrode(s)
- Reapply problematic electrode(s)

If participant has no sensation or mild, increase current amplitudes in both channels by ~10% of target amplitude.

tACS Contrast Test

Stimulation Parameters:

I1	
Frequency (Hz)	20
Amplitude (mA)	0.5
I2 OFF	
Frequency (Hz)	N/A
Amplitude (mA)	N/A

Make ramp-time sufficient to minimise sensation (~5s) and stimulation time long enough to record voltage/impedance (~5-10s).

Record the measurement voltage (target: $\ll$ Compliance Voltage)

Calculate the load Resistance from recorded voltage and set current amplitude (target: $< 5k\Omega$)

Ask Participant which applies (record the selection):

No Sensation
Mild sensation (tolerable)
Painful or Significant discomfort